C

IB MATHEMATICS SL • UNIT 2.01

Topic 2: Functions, Transformations & Modeling

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

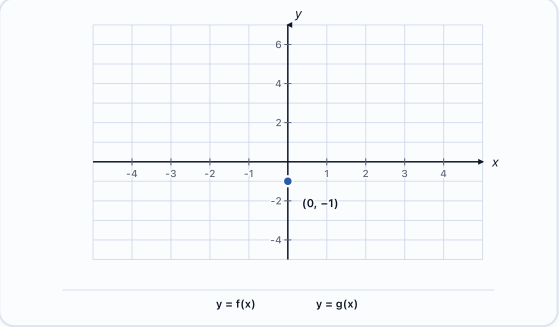
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 B	Q3 A	Q4 A	Q5 A
Q6 C	Q7 A	Q8 A	Q9 D	Q10 D
Q11 A	Q12 C	Q13 A	Q14 D	Q15 A
Q16 C	Q17 D	Q18 D	Q19 C	Q20 A
Q21 C	Q22 B	Q23 C	Q24 D	Q25 A

Topic 2: Functions, Transformations & Modeling Practice Exercises

1 MCQ

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. What are the coordinates of the y-intercept?



A $(0, -3)$ ✓

B $(0, 3)$

C $(-1, 0)$

D $(3, 0)$

WORKED SOLUTION & VERIFICATION

The y-intercept of a function occurs when $x = 0$. Substituting $x = 0$ into the function $f(x) = -x^2 + 4x - 3$ gives $f(0) = -(0)^2 + 4(0) - 3 = -3$. Therefore, the coordinates of the y-intercept are $(0, -3)$. **[M1] [A1]**

2 MCQ

Consider the function $f(x) = 2x + 3$. What is the value of $f(4)$?

A 7

B **11** ✓

C 10

D 5

WORKED SOLUTION & VERIFICATION

To find the value of $f(4)$, we substitute $x = 4$ into the function $f(x) = 2x + 3$.

$f(4) = 2(4) + 3$

$f(4) = 8 + 3$

$f(4) = 11$ **[M1] [A1]**

3 MCQ

Consider the function $f(x) = 2x - 3$. Find the value of $f(3)$.

- **A** 3 ✓
- **B** 6
- **C** 9
- **D** 0

WORKED SOLUTION & VERIFICATION

To find the value of $f(3)$, we substitute $x = 3$ into the function definition $f(x) = 2x - 3$.

$f(3) = 2(3) - 3$
$f(3) = 6 - 3$
$f(3) = 3$. **[M1] [A1]**

4 MCQ

Consider the function $f(x) = -x^2 + 6x - 5$. Determine the coordinates of the vertex of the graph of $y = f(x)$.

- **A** **(3, 4)** ✓
- **B** (-3, 4)
- **C** (3, -4)
- **D** (4, 3)

WORKED SOLUTION & VERIFICATION

The given function is a quadratic of the form $f(x) = ax^2 + bx + c$, where $a = -1$, $b = 6$, and $c = -5$. The x-coordinate of the vertex of a parabola is given by $x = -\frac{b}{2a}$. Substituting the values, we get $x = -\frac{6}{2(-1)} = -\frac{6}{-2} = 3$. To find the y-coordinate of the vertex, substitute $x = 3$ into the function: $f(3) = -(3)^2 + 6(3) - 5 = -9 + 18 - 5 = 9 - 5 = 4$. Therefore, the coordinates of the vertex are (3, 4). **[M1] [A1]**

5 MCQ

Consider the function $f(x) = 2x - 1$. Find the value of $f(3)$.

- **A** 5 ✓
- **B** 6
- **C** 7
- **D** 4

WORKED SOLUTION & VERIFICATION

$f(x) = 2x - 1$. To find $f(3)$, substitute $x = 3$ into the function: $f(3) = 2(3) - 1$. $f(3) = 6 - 1$. $f(3) = 5$. Therefore, the value of $f(3)$ is 5. **[M1] [A1]**

6 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown. Determine the value of $f(0)$.

- **A** 1
- **B** -1
- **C** 3 ✓
- **D** -3

WORKED SOLUTION & VERIFICATION

To find the value of $f(0)$, substitute $x = 0$ into the function $f(x) = x^2 - 4x + 3$. $f(0) = (0)^2 - 4(0) + 3$ $f(0) = 0 - 0 + 3$ $f(0) = 3$. Therefore, the value of $f(0)$ is 3. **[M1] [A1]**

7 MCQ

Consider the function $f(x) = -2x + 6$. The graph of $y = f(x)$ is shown below. Determine the value of the y-intercept.

- **A** 6 ✓
- **B** -3
- **C** 3
- **D** -6

WORKED SOLUTION & VERIFICATION

The y-intercept of a function is the value of y when $x = 0$. For the function $f(x) = -2x + 6$, we substitute $x = 0$ into the equation:

$f(0) = -2(0) + 6$

$f(0) = 0 + 6$

$f(0) = 6$

Therefore, the y-intercept is 6. **[M1] [A1]**

8 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown below. The function intersects the x-axis at two points. Find the coordinates of the point with the smaller x-value.

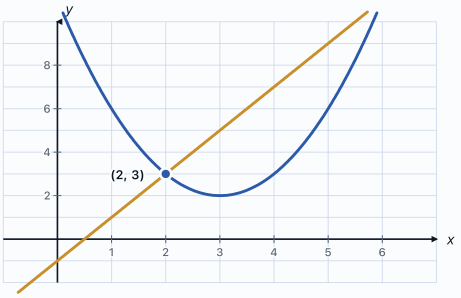
- **A** **(1, 0)** ✓
- **B** (3, 0)
- **C** (0, 3)
- **D** (0, 1)

WORKED SOLUTION & VERIFICATION

To find the x-intercepts of the function $f(x) = x^2 - 4x + 3$, we set $f(x) = 0$. This gives us the quadratic equation $x^2 - 4x + 3 = 0$. We can factor this equation as $(x - 1)(x - 3) = 0$. The solutions are $x = 1$ and $x = 3$. Therefore, the x-intercepts are at $(1, 0)$ and $(3, 0)$. The point with the smaller x-value is $(1, 0)$. **[M1] [A1]**

9 MCQ

Consider the functions $f(x) = x^2 - 6x + 11$ and $g(x) = 2x - 1$. The graphs of these functions intersect at two points. One of the intersection points is $(2, 3)$. Find the x-coordinate of the other intersection point.



A 3

B 4

C 5

D 6 ✓

WORKED SOLUTION & VERIFICATION

To find the intersection points of the two functions, we set $f(x) = g(x)$. $x^2 - 6x + 11 = 2x - 1$

Rearrange the equation to form a quadratic equation:

$x^2 - 6x - 2x + 11 + 1 = 0$

$x^2 - 8x + 12 = 0$

We can factor this quadratic equation. We are looking for two numbers that multiply to 12 and add to -8. These numbers are -6 and -2.

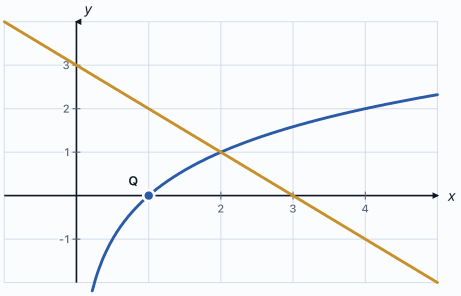
$(x - 2)(x - 6) = 0$

This gives us two possible solutions for x: $x = 2$ or $x = 6$.

We are given that one intersection point has an x-coordinate of 2. Therefore, the x-coordinate of the other intersection point is 6. **[M1] [A1]**

10 MCQ

Consider the two functions $f(x) = \log_2(x)$ and $g(x) = 3 - x$. The graphs of $f(x)$ and $g(x)$ intersect at a point P, and $f(x)$ intersects the x-axis at point Q. A line segment is drawn from Q to a point R on the graph of $g(x)$ such that QR is perpendicular to the x-axis. Find the coordinates of the point R.



A (4, -1)

B (3, 0)

C (1, 2)

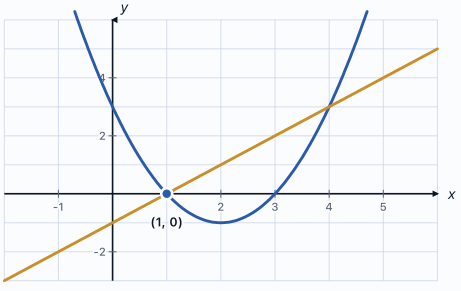
D **(2, 1)** ✓

WORKED SOLUTION & VERIFICATION

The graph of $f(x) = \log_2(x)$ intersects the x-axis when $f(x) = 0$. Thus, $\log_2(x) = 0$, which implies $x = 2^0 = 1$. So, point Q has coordinates (1, 0).
The line segment QR is perpendicular to the x-axis, meaning that the x-coordinate of R is the same as the x-coordinate of Q. Therefore, the x-coordinate of R is 1.
Point R lies on the graph of $g(x) = 3 - x$. Substituting the x-coordinate of R into the equation for $g(x)$, we get $g(1) = 3 - 1 = 2$. Thus, the coordinates of point R are (1, 2).
Let's check the intersection point P. The equation to solve is $\log_2(x) = 3 - x$. By inspection, when $x = 2$, $\log_2(2) = 1$ and $3 - 2 = 1$. So P is (2, 1). This information is not directly needed to find R, but confirms the graphs intersect.
[M1] [A1]

11 MCQ

Consider the function $f(x) = x^2 - 4x + 3$ and the line $g(x) = x - 1$. The graph of these functions is shown below. Find the coordinates of the point where the line intersects the parabola.



A **(1, 0)** ✓

B (3, 2)

C (2, 1)

D (4, 3)

WORKED SOLUTION & VERIFICATION

To find the intersection points, we set the two functions equal to each other: $f(x) = g(x)$ $x^2 - 4x + 3 = x - 1$ $x^2 - 5x + 4 = 0$ Factoring the quadratic equation: $(x - 1)(x - 4) = 0$ This gives us two possible x-values for the intersection points: $x = 1$ or $x = 4$. \nUsing $g(x) = x - 1$ to find the corresponding y-values: \nFor $x = 1$: $y = 1 - 1 = 0$. The intersection point is $(1, 0)$. \nFor $x = 4$: $y = 4 - 1 = 3$. The intersection point is $(4, 3)$. \nLooking at the graph, one of the intersection points is $(1, 0)$. **[M1] [A1]**

12 MCQ

Let $f(x) = 2x^2 + 6x - 5$. Find the y-intercept of the graph of f.

A -10	B 5
C -5 ✓	D -15

WORKED SOLUTION & VERIFICATION

Working through this functions step: The y-intercept occurs at $x = 0$: $f(0) = -5$. **[M1] [A1]**

13 MCQ

The point $(0, -3)$ lies on the graph of $y = f(x)$. Find the corresponding point on the graph of $y = f(x - 1) - 4$.

A $(1, -7)$ ✓	B $(-1, -7)$
C $(-4, -2)$	D $(1, 1)$

WORKED SOLUTION & VERIFICATION

Working through this functions step: Horizontal shift by 1 and vertical shift by -4: $(x, y) \to (x + 1, y - 4)$, giving $(1, -7)$. **[M1] [A1]**

14 MCQ

In this problem, the point $(4, -9)$ lies on the graph of $y = f(x)$. Find the corresponding point on the graph of $y = f(x - 1) - 2$.

A $(5, -7)$	B $(3, -11)$
C $(2, -8)$	D $(5, -11)$ ✓

WORKED SOLUTION & VERIFICATION

Working through this functions step: Horizontal shift by 1 and vertical shift by -2: $(x, y) \to (x + 1, y - 2)$, giving $(5, -11)$. **[M1] [A1]**

15 MCQ

Let $f(x) = x^2 + 2x - 9$. Find $f(3)$.

A 6 ✓	B 2
C 3	D 1

WORKED SOLUTION & VERIFICATION

Working through this functions step: $f(3) = (1)(3)^2 + (2)(3) + (-9) = 6$. **[M1] [A1]**

16 MCQ

The point $(-4, 1)$ lies on the graph of $y = f(x)$. Find the corresponding point on the graph of $y = -f(x)$.

A $(4, 1)$

B $(4, -1)$

C $(-4, -1)$ ✓

D $(-4, 1)$

WORKED SOLUTION & VERIFICATION

Working through this functions step: Reflection in the x-axis negates the y-coordinate: $(x, y) \to (x, -y)$, giving $(-4, -1)$. **[M1] [A1]**

17 MCQ

The point $(0, 7)$ lies on the graph of $y = f(x)$. Find the corresponding point on the graph of $y = -f(x)$.

A $(1, -7)$

B $(0, 7)$

C $(0, -6)$

D $(0, -7)$ ✓

WORKED SOLUTION & VERIFICATION

Working through this functions step: Reflection in the x-axis negates the y-coordinate: $(x, y) \to (x, -y)$, giving $(0, -7)$. **[M1] [A1]**

18 MCQ

The point $(-1, -7)$ lies on the graph of $y = f(x)$. Find the corresponding point on the graph of $y = f(x + 4) - 1$.

A $(3, -8)$

B $(-5, -6)$

C $(-2, -11)$

D $(-5, -8)$ ✓

WORKED SOLUTION & VERIFICATION

Working through this functions step: Horizontal shift by -4 and vertical shift by -1: $(x, y) \to (x - 4, y - 1)$, giving $(-5, -8)$. **[M1] [A1]**

19 MCQ

The point $(6, -2)$ lies on the graph of $y = f(x)$. Find the corresponding point on the graph of $y = f(x + 5) + 5$.

A $(11, -7)$

B $(1, -7)$

C $(1, 3)$ ✓

D $(11, 3)$

WORKED SOLUTION & VERIFICATION

Working through this functions step: Horizontal shift by -5 and vertical shift by 5: $(x, y) \to (x - 5, y + 5)$, giving $(1, 3)$. **[M1] [A1]**

20 MCQ

Let $f(x) = -x^2 + x - 6$. Find $f(2)$.

- **A** -8 ✓
- **B** -12
- **C** -18
- **D** 2

WORKED SOLUTION & VERIFICATION

Working through this functions step: $f(2) = (-1)(2)^2 + (1)(2) + (-6) = -8$. **[M1] [A1]**

21 MCQ

Let $f(x) = -2x^2 + 8x - 2$. Find the y-intercept of the graph of f.

- **A** 0
- **B** 3
- **C** -2 ✓
- **D** 1

WORKED SOLUTION & VERIFICATION

Working through this functions step: The y-intercept occurs at $x = 0$: $f(0) = -2$. **[M1] [A1]**

22 MCQ

Suppose we define $f(x) = -2x^2 + 7x - 1$. Find the y-intercept of the graph of f.

- **A** 0
- **B** -1 ✓
- **C** 1
- **D** 2

WORKED SOLUTION & VERIFICATION

Working through this functions step: The y-intercept occurs at $x = 0$: $f(0) = -1$. **[M1] [A1]**

23 MCQ

The point $(-3, -7)$ lies on the graph of $y = f(x)$. Find the corresponding point on the graph of $y = f(-x)$.

- **A** $(3, 7)$
- **B** $(-3, 7)$
- **C** $(3, -7)$ ✓
- **D** $(-3, -7)$

WORKED SOLUTION & VERIFICATION

Working through this functions step: Reflection in the y-axis negates the x-coordinate: $(x, y) \to (-x, y)$, giving $(3, -7)$. **[M1] [A1]**

24 MCQ

In this problem, the point $(1, 7)$ lies on the graph of $y = f(x)$. Find the corresponding point on the graph of $y = -f(x)$.

A $(-1, 7)$

B $(1, 7)$

C $(-1, -7)$

D $(1, -7)$ ✓

WORKED SOLUTION & VERIFICATION

Working through this functions step: Reflection in the x-axis negates the y-coordinate: $(x, y) \to (x, -y)$, giving $(1, -7)$. **[M1] [A1]**

25 MCQ

Let $f(x) = -x^2 - 7x - 4$. Find $f(-4)$.

A 8 ✓

B 3

C 4

D 5

WORKED SOLUTION & VERIFICATION

Working through this functions step: $f(-4) = (-1)(-4)^2 + (-7)(-4) + (-4) = 8$. **[M1] [A1]**